Lesson Planning

Complete Lesson Plan Book

Class: Class 10

Subject: Chemistry

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Acids and Bases: Properties and Uses

DM

Unit 1: Acids and Bases: Properties and Uses

Learning Objectives

• Students will describe the properties of acids and bases.

• Students will identify common acids and bases in everyday life.

• Students will explain neutralisation and name its products.

Key Concepts

• An acid is a substance that tastes sour, turns blue litmus red, and gives hydrogen ions in water.

• A base is a substance that feels slippery, turns red litmus blue, and gives hydroxide ions in water.

• Common acids include hydrochloric acid, sulphuric acid, citric acid in lemons and acetic acid in vinegar.

• Common bases include sodium hydroxide, calcium hydroxide and ammonia solution.

• When an acid reacts with a base, they neutralise each other and form a salt and water.

Teaching Methodology (Step by Step)

1. Ask students why lemon juice tastes sour and why soap feels slippery on the hands.

2. Define an acid and a base and write their key properties on the board.

3. Show the litmus test: blue litmus turns red in acid, red litmus turns blue in base.

4. Name common acids and bases with their everyday uses, such as vinegar and baking soda.

5. Explain neutralisation with the example of hydrochloric acid and sodium hydroxide forming salt and water.

6. Give students the sorting activity and summarise the properties in a table.

Classroom Activities

• Indicator test: students test lemon juice, soap water and plain water with litmus paper and record the colour change.

• Acid or base sort: students sort everyday items such as vinegar, baking soda solution and milk into acid and base groups.

• Neutralisation demo: the teacher adds a base to an acid drop by drop and students watch the indicator colour change.

Assessment Questions

1. What is the difference between an acid and a base?

2. Name two acids and two bases that we use in everyday life.

3. What happens when an acid and a base are mixed together?

Homework

• List five acids and five bases found at home or in the laboratory, with one use of each.

• Write the word equation for the neutralisation of hydrochloric acid with sodium hydroxide.

Time Allocation

40 minutes